# Business Performance Analysis Report

Data Analytics & Business Intelligence — Task 2 | Retail Sales Dataset (Orders + Customers)

## 1. KPI Overview

The combined Orders and Customers dataset (1,000 orders, 1,000 unique customers, Jan 2023 – Jan 2024) was joined on Customer_ID and analyzed across region, product category, and customer segment. Headline performance is summarized below.

| KPI | Value |
|---|---:|
| Total Sales | $456,000 |
| Total Profit | $71,306.28 |
| Overall Profit Margin | 15.64% |
| Total Orders | 1,000 |
| Total Customers | 1,000 |
| Average Order Value (AOV) | $456.00 |

## 2. Regional Performance

The North region generates the highest revenue at $133,380 (29.2% of total sales), followed by West ($113,075), South ($110,260), and East ($99,285), the lowest performer. North also holds the strongest profit margin (16.27%), meaning it leads on both volume and efficiency.

| Region | Total Sales | Total Profit | Profit Margin | Orders |
|---|---:|---:|---:|---:|
| North | $133,380 | $21,697.21 | 16.27% | 278 |
| West | $113,075 | $17,015.84 | 15.05% | 234 |
| South | $110,260 | $16,793.99 | 15.23% | 242 |
| East | $99,285 | $15,799.24 | 15.91% | 246 |

**Insight:** East has the fewest orders and the lowest total sales, but a healthy 15.91% margin — the gap versus North is a volume problem, not a profitability problem. Growth efforts in East should focus on order volume (marketing reach, customer acquisition) rather than pricing or discounting.

## 3. Category Performance

Beauty carries the highest profit margin by a wide margin (25.21%) despite having the lowest total sales of the three categories. Electronics generates the most revenue ($156,905) but converts it to profit far less efficiently (7.22% margin) — roughly a third of Beauty's rate.

| Category | Total Sales | Total Profit | Profit Margin | Orders |
|---|---:|---:|---:|---:|
| Electronics | $156,905 | $11,327.62 | 7.22% | 342 |
| Clothing | $155,580 | $23,804.99 | 15.30% | 351 |
| Beauty | $143,515 | $36,173.67 | 25.21% | 307 |

**Insight:** Electronics behaves like a volume/traffic driver rather than a margin driver. Beauty is the most profitable category per dollar of sales and is under-represented in total revenue relative to its profitability — a strong candidate for expanded marketing investment.

## 4. Customer Segment Analysis

| Segment | Total Sales | Total Profit | Profit Margin | Orders |
|---|---:|---:|---:|---:|
| Consumer | $224,925 | $34,750.84 | 15.45% | 506 |
| Corporate | $135,630 | $19,388.94 | 14.30% | 298 |
| Home Office | $95,445 | $17,166.50 | 17.99% | 196 |

**Insight:** Consumer is the largest segment by revenue (49.3% of total sales) and the backbone of the business, but Home Office is the most profitable per dollar sold (17.99% margin) despite being the smallest segment. "Most valuable" depends on the lens: Consumer for scale, Home Office for efficiency.

### Top 5 Customers by Revenue

| Rank | Customer | Region | Segment | Revenue |
|---|---:|---:|---:|---:|
| 1 | Daniel Miller | South | Corporate | $2,100 |
| 2 | Emily Miller | South | Corporate | $2,000 |
| 3 | Joshua Williams | West | Consumer | $2,000 |
| 4 | Patricia Brown | West | Consumer | $2,000 |
| 5 | John Gonzalez | South | Consumer | $2,000 |

## 5. Are Discounts Reducing Profitability?

Yes — clearly. Profit margin falls sharply as discount rate increases, and the correlation between discount and profit margin across all orders is -0.70, a strong negative relationship.

| Discount Band | Orders | Avg. Profit Margin | Total Sales |
|---|---:|---:|---:|
| 0% (no discount) | 309 | 24.63% | $135,245 |
| 1-10% | 446 | 16.03% | $208,945 |
| 11-20% | 208 | 4.44% | $94,500 |
| 20%+ | 37 | -1.12% | $17,310 |

**Insight:** Orders discounted above 20% are effectively unprofitable on average (-1.12% margin). Discounting past the 10% threshold destroys most of the margin gained from the sale. Deep discounts should be tightly controlled and reserved for clearing inventory, not used as a routine sales lever.

## 6. Is There a Seasonal Sales Trend?

Yes. Sales fluctuate meaningfully across the year rather than staying flat. May 2023 ($53,150) and October 2023 ($46,580) are peak months, while September 2023 ($23,620) is a sharp trough — less than half of the May peak. Quarterly totals confirm the pattern: Q4 ($126,190) and Q2 ($123,735) outperform Q1 ($110,030) and Q3 ($96,045), which lags noticeably behind the other three quarters.

**Insight:** Q3 (Jul–Sep) is consistently the weakest period and is a natural window for promotions or inventory clearance, while Q4 and Q2 are the strongest windows to prioritize stock availability and marketing spend.

## 7. Recommendations for Business Growth

- Cap routine discounts at 10%. Margins collapse past this point (16.03% → 4.44% → -1.12%); reserve deeper discounts for planned clearance events, not standard promotions.
- Invest further in Beauty. It has the highest margin (25.21%) but the lowest sales volume of the three categories — there is room to grow revenue without sacrificing efficiency.
- Review Electronics pricing and cost structure. At a 7.22% margin it is the least efficient category despite leading in revenue; even a small margin improvement here has an outsized profit impact given its volume.
- Grow order volume in the East region specifically. Its margin (15.91%) is healthy and close to North's; the shortfall versus North is driven by fewer orders (246 vs 278), not weaker unit economics.
- Protect and grow the Home Office segment. It is the smallest segment but the most profitable per dollar (17.99% margin) — a good candidate for targeted retention or loyalty offers.
- Plan inventory and marketing around the Q3 dip. Build promotional campaigns for July–September specifically to smooth out the seasonal trough, and ensure adequate stock ahead of the May and October peaks.

Prepared as part of Task 2 — Data Analytics & Business Intelligence internship program. Analysis based on SQL JOIN queries, pivot table analysis (Excel), and Power BI KPI dashboard built from the Orders and Customers datasets.